

Estimation of passion fruit yield based on YOLOv8n + OC-SORT + CRCM algorithm

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ABSTRACT

The precise estimation of passion fruit yield is crucial for efficient orchard management, but it poses challenges such as occlusion, light variations, and camera shake, which can lead to problems such as missed detection, error detection, and double counting of small fruits. In this study, we propose a robust computer vision algorithm named YOLOv8n + OC-SORT + CRCM (Central Region Counting Method) to accomplish three tasks: detection, tracking, and yield estimation of passion fruits. Firstly, we compare the passion fruit detection results using various YOLO series detection algorithms and choose YOLOv8n for detector. Then, OC-SORT algorithm is chosen as the tracker due to its effectiveness in addressing issues like occlusion, vertical shaking, and uneven speeds. Finally, we design CRCM counting algorithm for fruit counting for addressing challenges in estimating passion fruit yield. To validate the effectiveness of these methods, a real-world passion fruit video dataset including 24 videos for each with a 1-minute length was established. In the detection results on the test set, YOLOv8n detector achieved the best results with a mAP@0.5 (mean Average Precision) of 86.3 % and a model size of only 6.2 MB among YOLOv5n, YOLOv7 and YOLOv8n three detectors. Furthermore, the HOTA (higher order tracking accuracy) of OC-SORT tracker was 67.10 %, surpassing three mainstream trackers named BoT-SORT, Byte Track, and Strong SORT by 2.98 %, 4.71 %, and 8.82 %, respectively. In the fruit yield estimation, CRCM demonstrated an average counting accuracy of 87.0 %, surpassing ID number and Single Line Method (SLM) methods by 49.8 % and 10.5 %, respectively. In conclusion, the YOLOv8n + OC-SORT + CRCM algorithm effectively addresses issues of misidentification, missed detections of small fruits, and repeated counts, achieving stable, real-time, and accurate estimation of passion fruit yield.

1. Introduction

Passion fruit pulp has great potential for the development of different functional products and presents broad prospects in maintaining health (Pereira et al., 2023). The passion fruit is cultivated primarily in tropical and subtropical regions, which stands as a significant indigenous crop in South America, holding substantial economic value. Accurate estimation of passion fruit yield is crucial for farmers, aiding them in more effectively planning cultivation areas, selecting harvest times, and devising sales strategies. However, traditional manual estimation methods are tedious and time-consuming. In recent years, the widespread application of computer vision technology has offered effective support in fruit counting, presenting new possibilities for automating passion fruit yield estimation (Farjon et al., 2023; He, Fang, et al., 2022; Vasconez et al., 2020).

In fruit yield management, numerous researchers employed multiple

object tracking (MOT) algorithms such as simple online and real-time tracking (SORT) or DeepSORT for fruit counting and achieved remarkable results (Gao et al., 2022; Tan et al., 2022; Villacrés et al., 2023). The MOT algorithms can implement the count of fruits by locating the fruit objects in the video and assigning a unique ID to each detected object (Bashar et al., 2022; Guo et al., 2022). For instance, Leiying He et al. (2022) utilized YOLO-v3 detection results, coupling detection boxes with trajectories using Cascade- MOT algorithm, enhancing the accuracy and robustness of fruit counting. Egi et al. (2022) employed YOLOv5 detector and DeepSORT tracker for counting tomato flowers and fruits. Yang et al. (2022) developed an advanced anchor-free object detection model using CenterNet to detect cotton seedlings and extract their identity embeddings, employing DeepSORT for positional and identity-based data association. The above proposed many fruit detection or tracking algorithms have enhanced the performance of automated fruit counting systems and improved the accuracy of fruit yield prediction in

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orchards.

However, tracking fruits in real-world scenarios often faces challenges such as occlusion and varying lighting conditions, leading to missed or false detection and frequent ID switches (Miranda et al., 2023). Improvements are generally pursued either through enhancing the detection efficacy of the detection model or the stability of tracking algorithms (Li et al., 2023; Xie et al., 2023). In terms of refining detection performance, Ge et al. (2022) introduced a visual object tracking network named YOLO-DeepSORT. Leveraging the YOLOv5s model with shufflenetv2 and CBAM attention mechanism, this approach aims to enhance detection accuracy. Furthermore, Wu et al. (2023) proposed an automated fruit counting method known as NDMFCS (Normal Detection Matched Fruit Counting System). This method defines fruits behind leaves and those on the ground as abnormal fruits and excludes them during the detection process to enhance counting accuracy. Concerning the stability of tracking algorithms, Farjon et al. (2023) proposed a motion displacement estimation algorithm OrangeSort based on the SORT algorithm to significantly reduce the error associated with the repetitive counting problem. Gao et al. (2022) chose tree trunks as single-target tracking targets for apple tracking to achieve higher accuracy and faster tracking speed than the commonly used fruit-based multi-target tracking methods. Our previous work focused on passion fruit yield counting using lightweight YOLOv5s and improved DeepSORT (Tu et al., 2024). In fruit yield management, with the emergence of more and more outstanding MOT algorithms, the fruit yield estimation is gaining increasingly in accuracy under natural scenarios.

At present, many studies estimate fruit quantity based on the number of IDs obtained from MOT algorithms. However, due to frequent ID switches, the resulting count often surpasses the actual number of fruits. Consequently, scholars have attempted to mitigate this issue by implementing various counting methods during the tracking process. Parico and Ahamed. (2021) compared ROI (Region of Interest) and unique ID methods. The ROI method counts fruits based on the number of tracked objects crossing the horizontal line, and the experimental results show that the fruit counts from the ROI method are closer to the artificial true counts. Rong et al. (2023) devised a specific tracking area counting method to overcome the problem of tracked tomato cluster ID switches. Zheng et al. (2023) proposed a nonuniform distributed counter to rectify citrus fruit counting during tracking, aiming to reduce counting errors resulting. In some fruit yield estimation, it may be inaccurate to adopt max IDs counted as the yield result for frequent ID error switches. Therefore, there is an urgent demand for researchers to study the different types of fruit yield counting algorithms according to the results of MOT.

Despite the widespread application of MOT methods in fruit counting and numerous improvements made specifically for fruit counting, there has yet to be comprehensive research that systematically compares and summarizes the mainstream detectors, tracking algorithms, and counting methods. This study aims to address common challenges in estimating passion fruit yield, including factors such as variations in lighting, background interference, camera angles, and camera shaking, to identify the optimal detector, tracker, and counting method, thereby determining the best strategy for this task. Specifically, the main objectives of this research include the following three points:

- (1) A comprehensive comparison of the current mainstream YOLO series detection algorithms is conducted to obtain optimal detection performance. And we ultimately select YOLOv8n algorithm as the detector for this task.
- (2) Mainstream fruit tracking algorithms, including OC-SORT, BoT-SORT, Byte Track and Strong SORT trackers, are compared and evaluated for passion fruit tracking. The best performing OC-SORT tracker is finally selected.
- (3) A comprehensive comparison and evaluation of three commonly used counting methods, namely maximum ID number, single line method (SLM), and central region counting method (CRCM), are

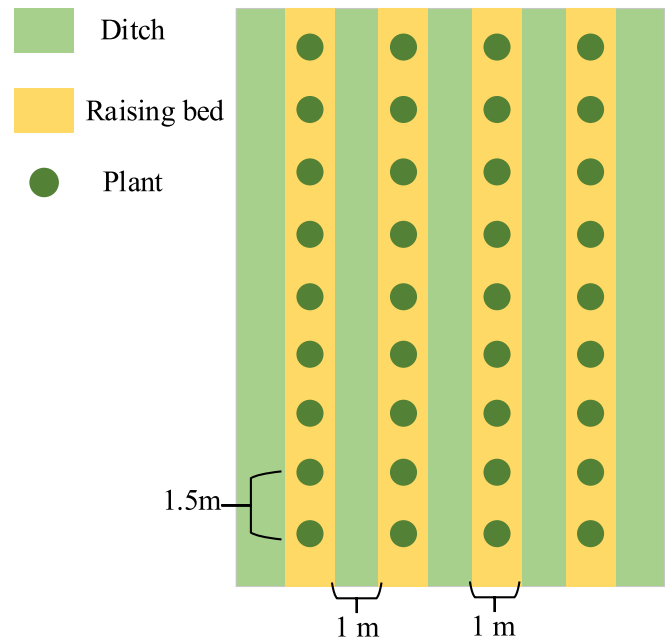


Fig. 1. Passion fruit cultivation.

performed to determine the most suitable strategy for estimation of passion fruit yield. And CRCM is selected as the best appropriate passion fruit yield estimation method among three approaches.

2. Dataset description

2.1. Dataset collection

To evaluate the effectiveness of our approach, we collected the videos dataset in passion fruit orchards. These videos were captured on October 14, 2022, under clear weather conditions in a passion fruit plantation located in Zengcheng District, Guangzhou City, Guangdong Province, China. They employed the Ridge Cultivation Method to cultivate passion fruit, with a spacing of 1.5 m between plants, 1 m width for ditches, and a raising bed width of 1 m. Specific details regarding the layout of the passion fruit plantation are shown in Fig. 1. We used the mobile phones for the video shoot with walking along ditch, the angle of mobile phone was vertical with the passion fruit raising bed, and the distance of the phone from the fruit was 1 m and the height was 1.5 m.

The dataset comprises a total of 24 videos, all captured by individuals using smartphones, with each video lasting approximately 1 min. These videos have a resolution of 1080*1920 and a frame rate of 30 frames per second. They are divided into two groups: half were shot away from the sunlight, while the other half were shot toward the sunlight. We were able to capture clear footage of the fruits under favorable lighting conditions and appropriate angles, as depicted in Fig. 2(a). However, we found the following issues during the collection of the dataset: (1) the image would suddenly become dark when shooting toward the sunlight, as illustrated in Fig. 2(b). (2) the picture will be blurred when the camera moves quickly, as shown in Fig. 2(c). The ratio of images taken towards and away from the sun is 5:5. And the image taken by camera shake is about 200 images from the total volume of frames. We conducted experiments and analyzed results using these datasets to evaluation the effectiveness of our approach.

2.2. Data processing

The processing flow of the dataset is shown in Fig. 3. From all 24



Fig. 2. Part of the dataset.

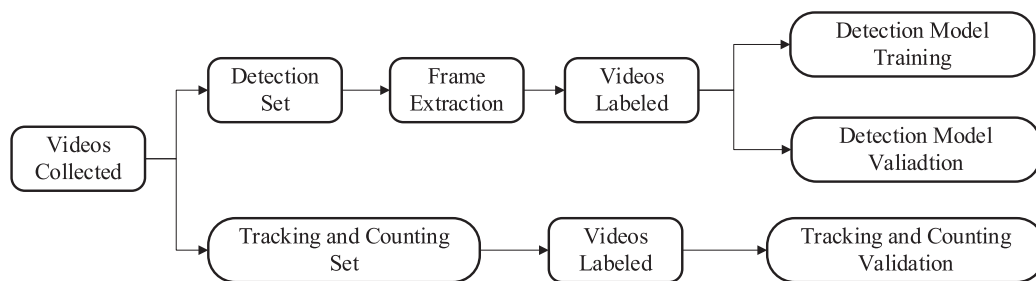


Fig. 3. Processing flow of the dataset.

videos, 5 videos were randomly selected for verifying the performance of the detection model and for the target tracking and counting tasks.

The processing of the detection set began with frame extraction, adjusting the video frame rate to 5 frames per second. And the videos were labeled using the DrakLabel software and cut into image sequences using the FFmpeg software. The labelled files of the videos were then converted into labelled file in YOLO and COCO format for the subsequent detection model training and validation process. Finally, all images and their corresponding labelled files were organized according to the division of the training and validation set.

For the five videos in tracking and counting set, they were directly labeled using DrakLabel software to be used as validation data for these tracking and counting tasks.

3. Methods

3.1. Method process and flow chart

We propose a passion fruit yield estimation algorithm based on YOLOv8n + OC-SORT + CRCM, the flow of which is shown in Fig. 4. Initially, YOLOv8n is used to detect each passion fruit and obtain its detection box and confidence in each frame of the video. Subsequently,

the detection result is fed into OC-SORT tracker, where the tracking information of each passion fruit is achieved through three modules: observation-centered updating strategy, data association, and tracklet management. After the tracking process, ID number, SLM and CRCM are used to count passion fruits for yield estimation. Finally, the MOT visualization results are output, along with a text message of the passion fruit counting results in the upper left corner of each frame.

3.2. YOLOv8

The object detection algorithms of YOLO series employ a single-stage detection approach. This avoids the candidate box generation and filtering steps in traditional two-stage detection methods, thereby simplifying the algorithmic process (Redmon et al., 2018; Bochkovskiy et al., 2020). Over the years, the YOLO algorithm has undergone significant improvements, with multiple releases that have continually improved its efficiency.

YOLOv8 was released by Ultralytics in January 2023 (Talaat & ZainEldin, 2023). Compared to its predecessors, it brings a series of improvements: closing the Mosaic Augmentation, adoption of new convolutional modules, and anchor free detection. Firstly, YOLOv8 closes mosaic augmentation in the final ten training epochs. This

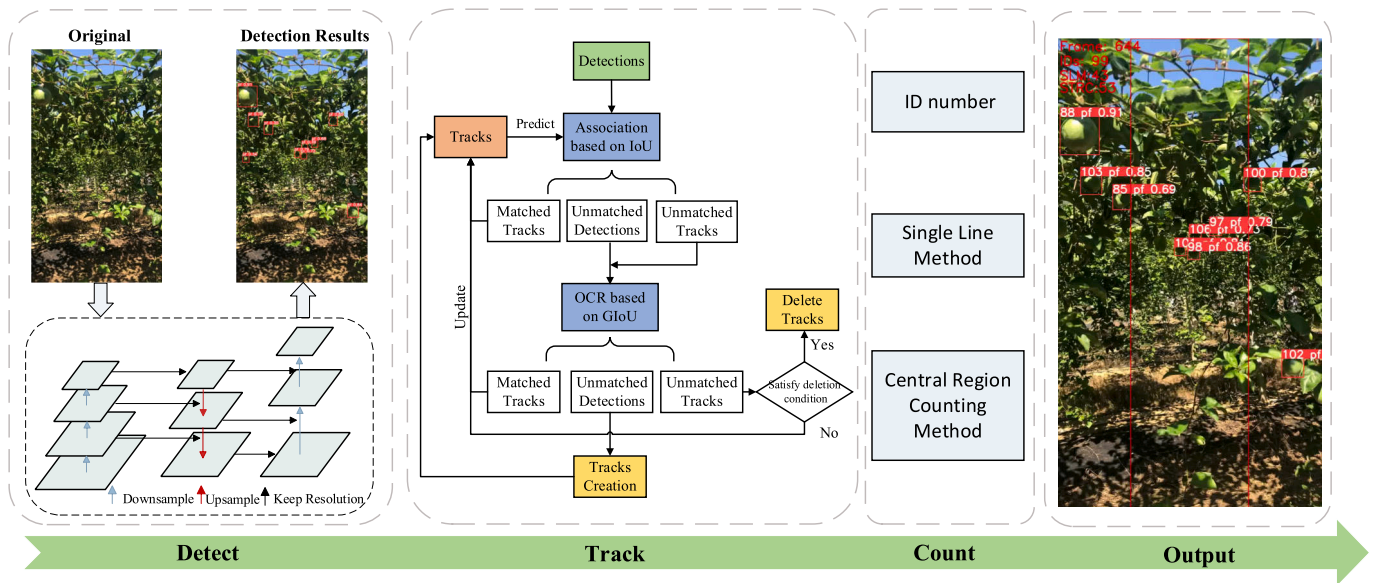


Fig. 4. Passion fruit yield estimation algorithm flow chart based on YOLOv8n + OC-SORT + CRCM.

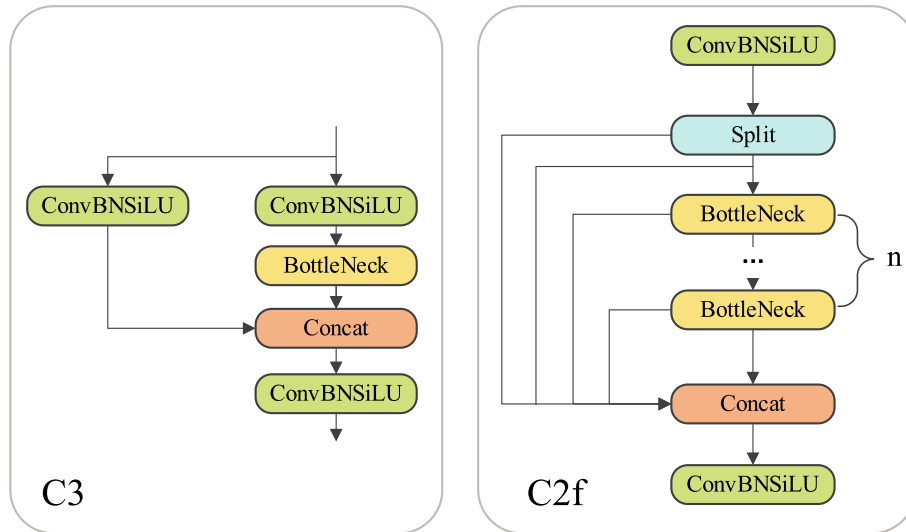


Fig. 5. C3 module and C2f module.

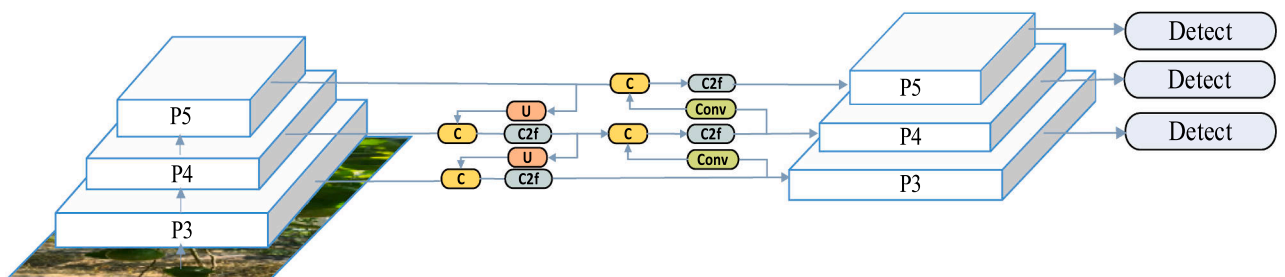


Fig. 6. YOLOv8 Architecture.

improvement is motivated by the observation that using this enhancement throughout the entire training process could decrease performance. Secondly, YOLOv8 replaces the C3 module with the C2f module.

Fig. 5 illustrates the structures of the C3 and C2f modules. The C2f module is designed based on the C3 module and the efficient layer aggregation networks, allowing YOLOv8 to acquire richer gradient flow

information while being lightweight. Lastly, anchor-free detection is employed. This approach directly predicts the object's center instead of relying on offsets based on known anchor boxes. This improvement effectively reduces the number of candidate detections, thus speeding up the post-processing step of non-maximum suppression.

Fig. 6 illustrates the structure of YOLOv8. The YOLOv8 model has

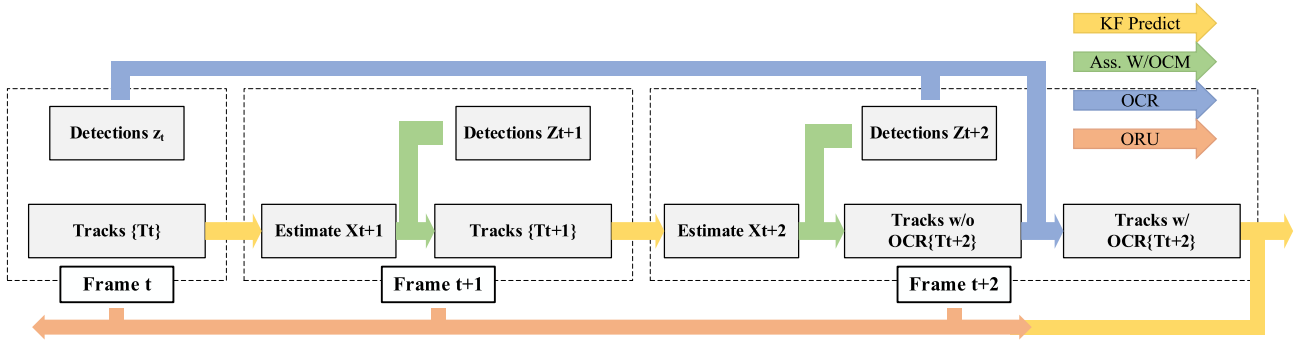


Fig. 7. Pipeline of OC-SORT.

completed structural adjustments, modifying the model's backbone, key building blocks, and fusion layers to render the model more compact and efficient. The backbone network employs the spatial pyramid pooling fast (SPPF) module, while the neck network retains the path aggregation network (PAN)-feature pyramid network (FPN) methods for feature fusion. In the detection head section, an anchor-free detection head is utilized. YOLOv8 was selected as the primary detector to meet the real-time and accurate detection requirements for passion fruit in practical scenarios.

3.3. OC-SORT

Currently, most target tracking methods rely heavily on the Kalman filter(KF), which employs a center-based estimation update strategy (Bewley et al., 2016; Wojke et al., 2017). These tracking algorithms update matched tracks without observation and put complete trust in the estimation for tracks. This may make it subject to severe noise when it suffers from occlusion or when the object motion is not completely linear. In fruit yield prediction work, the KF tracking algorithm achieved

good performance under unconcluded, or slightly occluded and linear motion scenarios for not the long videos. However, in the passion fruit dense and target occluded scenario, the KF tracking algorithm may affect the tracking performance due to the existence of a certain amount of noise and the cumulative error in target tracking caused by long time target occlusion.

During the data collection process, we employed a handheld method for capturing images with a mobile device for convenience. This method easily introduces vertical shaking and uneven speeds, resulting in non-linear motions. Additionally, we observed instances where targets were occluded by leaves during the shooting process. Therefore, if we used the KF tracking algorithm to track passion fruit in heavily obscured scenes, the tracking results may be moderately affected.

To address the above issues, OC-SORT introduces three key improvements: Observation-Centric Re-Update (ORU), Observation-Centric Momentum (OCM), and Observation-Centric Recovery (OCR) (Cao et al., 2023). These enhancements aim to boost tracking performance in scenarios involving occlusion and non-linear object motion.

ORU: When a trajectory is re-associated with observations after a

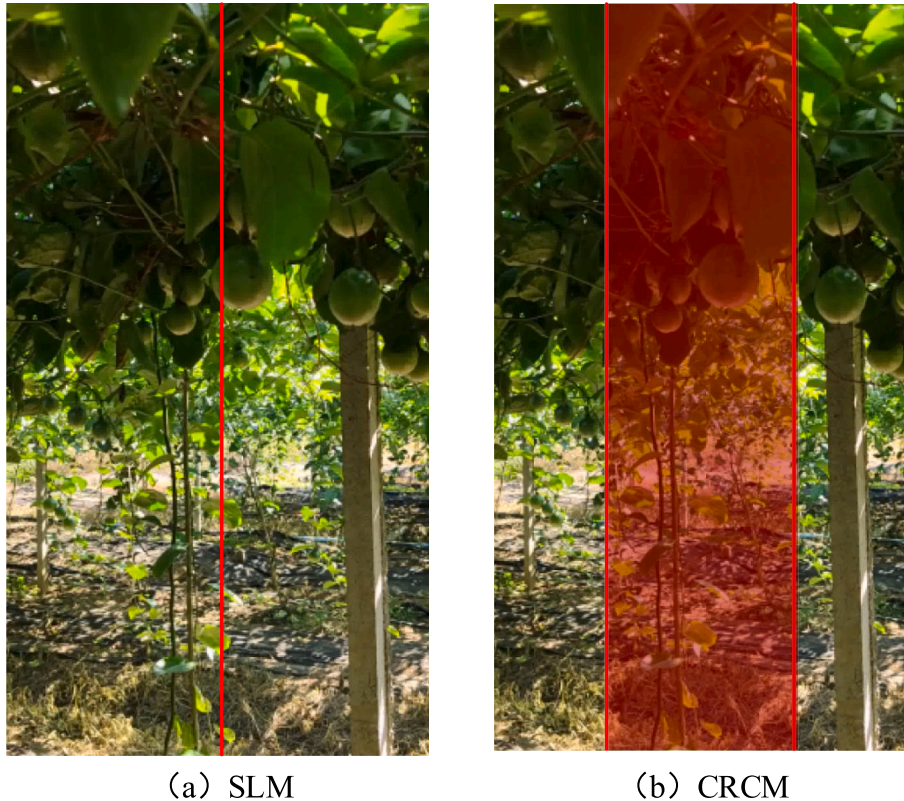


Fig. 8. Counting areas of SLM and CRCM.

period of being untracked, ORU strategy is employed to mitigate the accumulated errors during this untracked period. Initially, when generating a virtual trajectory, considerations are made regarding the observations at the beginning and end of the untracked period. The calculation for the virtual trajectory is represented by Equation (1).

$$\tilde{Z}_t = \text{traj}_{\text{virtual}}(z_{t_1}, z_{t_2}, t), t_1 < t < t_2 \quad (1)$$

where z_{t_1} corresponds to the last observation before the object was untracked, and z_{t_2} represents the observation when the object is re-associated.

Subsequently, the parameters of the Kalman filter are readjusted based on the virtual trajectory, as outlined in Equation (2).

$$\text{re-update} \begin{cases} K_t = P_{t|t-1} H_t^T (H_t P_{t|t-1} H_t^T + R_t)^{-1} \\ \hat{X}_{t|t} = \hat{X}_{t|t-1} + K_t (\tilde{z}_t - H_t \hat{X}_{t|t-1}) \\ P_{t|t} = (I - K_t H_t) P_{t|t-1} \end{cases} \quad (2)$$

OCM: it uses observations to compute the direction of motion of a trajectory as part of the cost matrix. It aims to increase the directional consistency of trajectories and thus reduce erroneous associations

OCR: it adds a second round of GIoU-based correlation after the first round of correlation to improve the tracking accuracy.

Fig. 7 shows the flowchart of OC-SORT tracker. If any target is lost on frame $t + 1$ but on the next frame, it is recovered by referring to the OCR's observation of frame t . The OCR's observation of frame t is used as a reference for the recovery of the target. Retracing triggers an ORU of its KF parameter from frame t to frame $t + 2$.

3.4. Existing counting algorithms for yield estimation

The three methods, ID number, SLM, and CRCM were employed to count passion fruits for yield estimation, and we designed the CRCM approach for passion fruit planting scenes based on the specific tracking area counting (Rong et al., 2023).

- (1) The ID number Counting method is relatively direct and simple, estimating the total number of fruits using the quantity of tracklets obtained from tracking algorithms.
- (2) The SLM method counts passion fruits based on whether they cross predetermined vertical lines. Specifically, when a passion fruit is tracked crossing the central vertical line, the total fruit count increases by 1. In the experiment, the starting coordinates of the line are (540, 0), and the ending coordinates are (540, 1920), as illustrated in Fig. 8(a).
- (3) The CRCM counts passion fruits based on whether the centroid of a passion fruit enters a predefined counting area. This method contains two counting rules: firstly, if a passion fruit enters the counting area from outside, the total count increases by 1; secondly, if a passion fruit that was previously in the counting area exits the area in the current frame, the total is added by 1. In this experiment, the coordinates of the upper left and lower right corners of the counting area are (300, 0) and (780, 1920), respectively as illustrated in Fig. 8(b).

4. Experiment and result analysis

4.1. Implementation details

The passion fruit yield estimation experiments in this study were divided into three parts. (1) Detection of passion fruit using YOLOv5s, YOLOv5m, YOLOv7-tiny(Wang et al., 2023), YOLOv8n and YOLOv8s models. (2) Passion fruit tracking using OC-SORT, BoT-SORT, ByteTrack, Deep OC-SORT, Strong SORT based on YOLOv8n model. (3) Passion fruit counting using the number of IDs, SLM and CRCM based on

Table 1

The detection results based on different detectors.

Model	P/%↑	R/%↑	F1/%↑	mAP@.5/%↑	Size/MB↓
YOLOv5s	81.3	79.7	80.5	86.4	14.4
YOLOv5m	79.9	80.6	80.3	86.5	42.5
YOLOv7-tiny	81.0	80.9	81.0	87.1	12.0
YOLOv8n	80.4	80.4	80.4	86.3	6.2
YOLOv8s	80.4	81.2	80.8	86.9	22.5

YOLOv8n and OC-SORT.

In detection experiments, the parameters are set as follows: epoch is 200, batch size is 32, input image size is 640, learning rate is 0.01, optimizer is SGD, conf-thres is 0.25, and iou-thres is 0.45. We used migration to train the YOLOv8n model, merging the standard categories into a new category 'passion fruit'. In total, 6445 images were used for detection, of which 5051 were used as the training set and 1394 as the validation set. And we used the data enhancement operations including Mosaic, Mixup, random perspective and HSV augment. In the tracking experiments, the parameters used are the default parameters in each tracking algorithm. In the counting experiments, the start coordinates of the vertical line were set to (540, 0) and the end coordinates were set to (540, 1920) in SLC. CRCM sets the coordinates of the upper left corner of the region to (300, 0) and the lower right corner to (780, 1920).

4.2. Evaluation metrics

The evaluation of the target detection section involves assessing key metrics such as precision (P), recall (R), F-Measure (F1), mean average precision (mAP), and model size for evaluating detector performance. Precision (P), Recall (R), F1 and mAP collectively indicate the detector's overall performance(Padilla et al., 2020), with higher values signifying better performance. Model size serves as indicators of the detector's efficiency, with lower values denoting greater efficiency.

The MOT segment is evaluated using higher order tracking accuracy (HOTA), multiple object tracking accuracy (MOTA), identification F1 (IDF1), and ID switch (IDSW) (Luiten et al., 2021). HOTA, MOTA, IDF1, and IDSW are crucial metrics for assessing tracking algorithm performance, where higher values of HOTA, MOTA, and IDF1 indicate superior performance, while a smaller value of IDSW signifies better tracking accuracy.

For the counting component, the evaluation relied on counting accuracy and statistical average counting accuracy(Rong et al., 2023). Higher values of these metrics suggest that the counting method's predictions are closer to the true values.

4.3. The detection results based on different detectors

Table 1 presents a comprehensive evaluation of different models in detecting passion fruits in practical scenarios. These evaluations include metrics such as precision, recall, F1, and mAP@0.5. The performance of each model is quite similar, with an average precision of 80.6 %, an average recall of 80.56 %, an average F1 of 80.58 %, and an average mAP@0.5 of 86.64 %. However, when considering model size, significant differences can be observed among them. The YOLOv8n model is only half the size of the best-performing YOLOv7-tiny model and had similar detection performance. This means that the YOLOv8n model maintains high accuracy while having a smaller model size, making it more suitable for online, real-time applications, and deployment on portable devices. Therefore, considering the balance of accuracy and capacity, we chose to use the YOLOv8n model as the detector in the subsequent tracking experiments.

Fig. 9 showed the detection results of different detectors in a practical scenario. In the scenario, there were a total of 24 passion fruits, primarily concentrated in the upper part of the image, presenting a higher density. There were fruits overlapping, and some fruits were

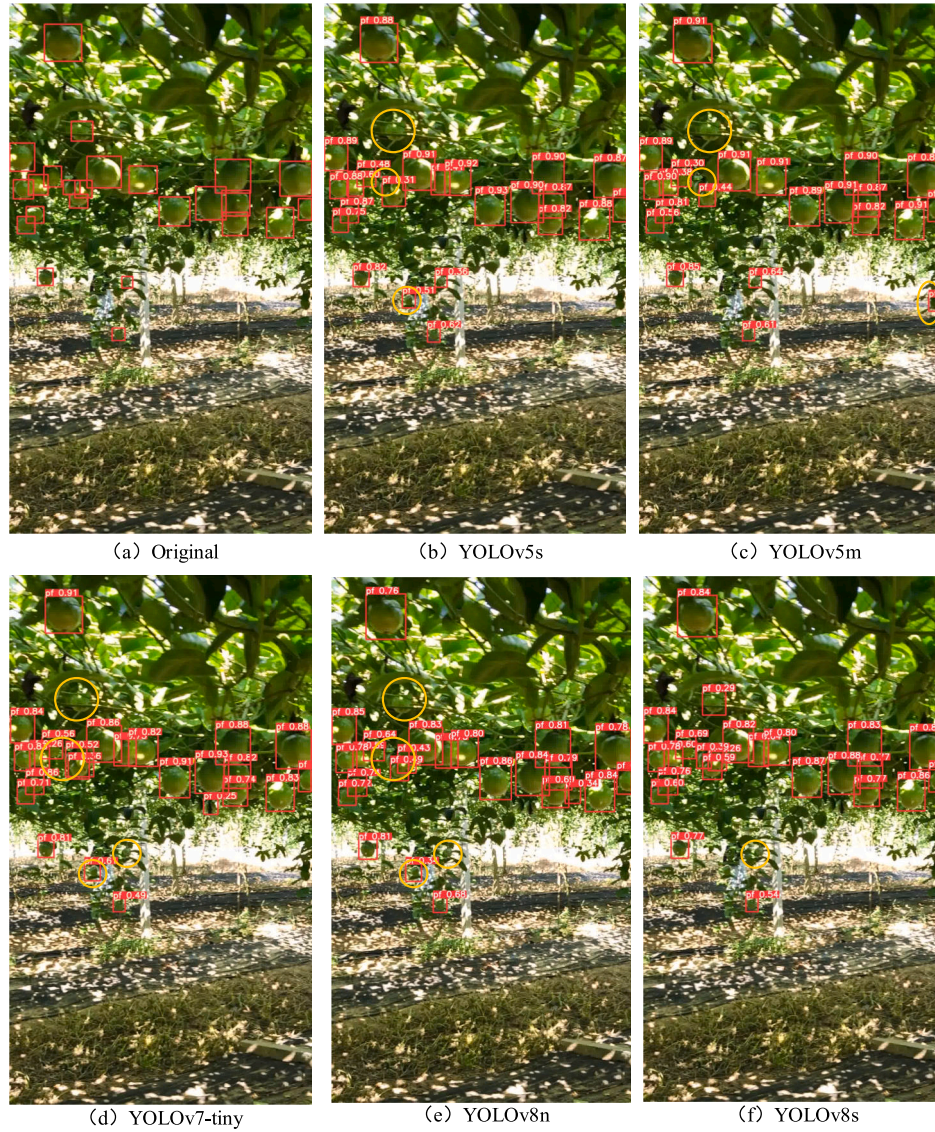


Fig. 9. The detection results of different detectors in practical scenario.

Table 2

The Tracking results based on different tracking algorithms.

Method	HOTA/% $\uparrow$	MOTA/% $\uparrow$	IDF1/% $\uparrow$	IDSW $\downarrow$	Time/ms $\downarrow$
OC-SORT	67.103	75.874	75.618	354	19.0
BoT-SORT	64.126	74.782	71.889	716	106.6
Byte Track	62.396	72.429	74.644	264	15.4
Strong SORT	58.280	71.864	61.639	578	87.3

partially obscured by leaves. All detection models could detect most of the passion fruits in this scenario, but misdetections and missed detections were still present, which were marked with yellow boxes. Firstly, YOLOv5s, YOLOv5m, YOLOv7-tiny, and YOLOv8n did not detect the fruit in the upper left due to leaf occlusion. Secondly, on the left side of the central fruit-dense area, the fruits overlapped each other, resulting in under-detection by the models YOLOv5s, YOLOv5m, YOLOv7-tiny, and YOLOv8n. Finally, in the lower half of the image, all detection models missed or misidentified leaves as fruits due to the relatively small size of the fruits and their similar shape to leaves. When we evaluated the detection performance of passion fruits, we found that detection accuracy was influenced by various factors, including environmental lighting, the similarity between leaves and fruits, occlusion issues, and

the small size of the fruits.

4.4. The tracking results based on different tracking algorithms

Table 2 presents a comparison of tracking results in test videos for OC-SORT, BoT-SORT(Aharon et al., 2022), Byte Track(Zhang et al., 2022), and Strong SORT(Du et al., 2023). From the perspectives of HOTA, MOTA, and IDF1, OC-SORT stands out with scores of 67.103 %, 75.874 %, and 75.618 %, respectively. Compared to BoT-SORT, Byte Track, and Strong SORT, OC-SORT exhibited improvements in HOTA by 2.977 %, 4.707 %, and 8.823 %, MOTA by 1.092 %, 3.445 %, and 4.01 %, and IDF1 by 3.729 %, 0.974 %, and 13.979 %, respectively. Furthermore, in terms of IDSW and runtime, Byte Track performed exceptionally well with an IDSW of 264 and an average runtime of 15.4 ms per frame. Taking all these metrics into consideration, OC-SORT shows significant advantages in comprehensive tracking metrics HOTA, MOTA, and IDF1, even though OC-SORT has a slight gap with Byte Track in IDSW and runtime. This demonstrates that OC-SORT possesses a unique competitive advantage in addressing the association and identification of passion fruits in practical scenarios.

Fig. 10 illustrated the visual results of target tracking in video 41 for OC-SORT, BoT-SORT, Byte Track, and Strong SORT. According to the

A.OC-SORT



B.BoT-SORT



C.Byte Track



D.Strong SORT



3rd frame in video 41

303rd frame in video 41

603rd frame in video 41

Fig. 10. Comparison of tracking results for OC-SORT, BoT-SORT, ByteTrack and Strong SORT in video 41.

Table 3

Tracking results based on YOLOv8n and OC-SORT on all test videos.

Test Videos	Fruit Number	HOTA/ %↑	MOTA/ %↑	IDF1/ %↑	IDSW↓	Time/ ms↓
10	136	65.776	78.811	81.848	12	17.6
21	329	67.412	77.592	75.360	65	21.4
41	203	70.081	72.808	77.969	97	17.4
61	348	63.365	72.659	71.201	128	18.5
90	188	67.475	78.845	74.612	52	20.2
All	1204	67.103	75.874	75.618	354	19.0

manually labeled ground truth file of video 41, the maximum IDs of the first frame, 301st frame, and 601st frame were 6, 42, and 120, respectively. At frame 3, the maximum IDs of all tracking algorithms were the same as the maximum ID in the ground truth. The differences in the visualization results of the algorithms were mainly found in frames 303 and 603. At frame 303, OC-SORT, BoT-SORT, Byte Track, and Strong SORT had maximum IDs differing from the maximum ID in the ground truth file by 4, 20, 5, and 24, respectively. These differences further increased in the 601st frame. OC-SORT, BoT-SORT, Byte Track, and Strong SORT had maximum IDs differing from the GT file by 20, 35, 39,

and 44, respectively. From the above results, the maximum ID value obtained by OC-SORT was closer to the maximum ID value of the GT file than other tracking algorithms, indicating that it had stability in the continuous tracking of passion fruit. Based on the tracking evaluation index and visualization results, we decided to adopt the OC-SORT model as our main tracking algorithm in the subsequent tracking experiments.

4.5. Tracking results based on YOLOv8n and OC-SORT on all test videos

Table 3 presented the tracking results of all test videos based on YOLOv8n and OC-SORT. While the variation in the number of fruits did not seem to directly impact the tracking performance of the videos, there were discernible patterns when classifying videos based on shooting conditions. Videos shot away from the sunlight (test videos 21, 41, 61) showed a decrease in tracking performance with an increase in the number of fruits. On the other hand, videos shot toward the sunlight (test videos 10, 90) exhibited a decrease in IDF1 and IDSW with an increase in fruit quantity. Moreover, despite test video 41 having a higher number of fruits compared to videos shot in sunlight, it displayed the best tracking performance. This suggested that the shooting conditions significantly influenced the tracking performance. In summary, these

**Fig. 11.** OC-SORT tracking results on test video 10, 21.

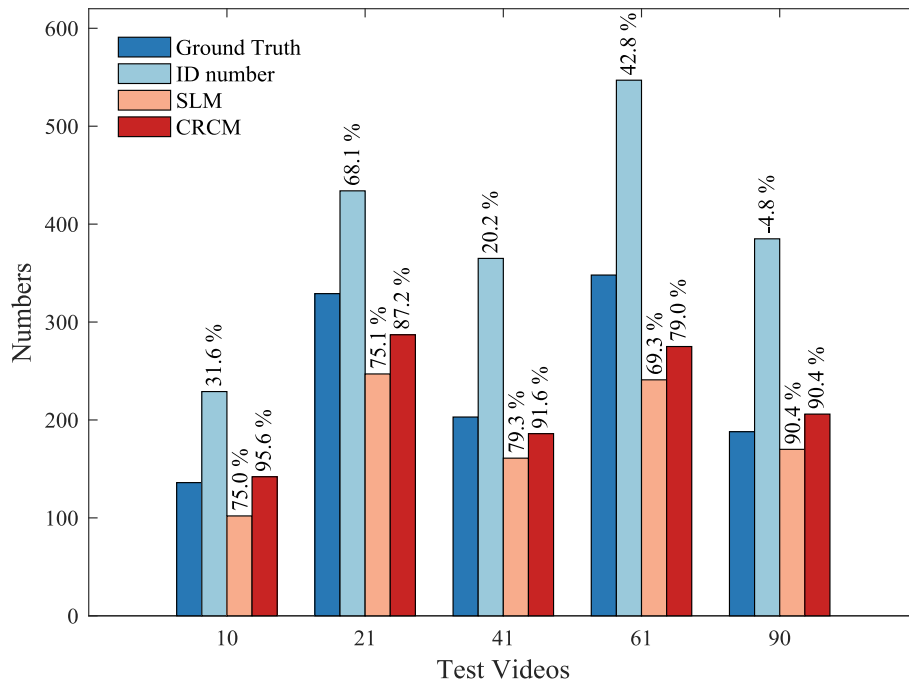


Fig. 12. The comparison results of the ground truth and the predicted fruit counting methods including ID number, SLM, and CRCM.

results implied that high fruit quantity and shooting toward the sunlight might have posed more challenges for object detection and tracking algorithms, such as increased interference between targets, mutual occlusion, and problems with frame exposure.

Fig. 11 presented the visual results based on OC-SORT in test videos 10 and 21. We studied the test videos for two different shooting conditions. Test video 10 was shot toward the sunlight, exhibiting occlusion of the leaves and darker scenes. Test video 21 was shot away from the sunlight, with certain areas showcasing densely packed and smaller-shaped fruits. OC-SORT tracked most of the fruits in both test videos 10 and 21. However, some fruits in the scene could not be tracked due to occlusion or inadequate lighting conditions. Fig. 11(a) to Fig. 11(c) exhibited the visual results for test video 10, there were untracked fruits in all three selected frames, due to leaf occlusion and low light issues. Conversely, Fig. 11(d) to Fig. 11(f) displayed the visual results for test video 21, with only one fruit not successfully tracked. The comprehensive analysis showed that the main reason for the poor tracking results was that the fruits in the scene were easily missed and the leaves were misidentified as fruits.

4.6. Evaluation of counting methods

Fig. 12 showed the comparison results of the ground truth and the predicted fruit counting methods including ID number, SLM, and CRCM. The x-axis represented the information for the test videos, while the y-axis represented the fruit counts. The text labels on the bar graphs were the counting accuracy of each prediction method for the test videos. Based on the results in Fig. 12, we found that CRCM method obtained highest counting accuracy than ID number and SLM in most videos. When compared to the SLM method, the CRCM improved the counting accuracy in each test video by 64 %, 19.1 %, 71.4 %, 36.2 %, and 95.2 %, respectively. When compared to the SLM method, the CRCM method exhibited counting accuracy improvements of 20.6 %, 12.1 %, 12.3 %, and 9.7 % in the first four test videos. Moreover, the average counting accuracy of ID number, SLM, and CRCM was 37.2 %, 76.5 %, and 87.0 %, respectively. In terms of counting accuracy and average counting accuracy, CRCM outperformed the ID number and SLM.

Fig. 13 presented the visualization results of the three different

counting methods. Fig. 13(a)- Fig. 13(c) showed the visualization results of SLM, the counting results of ID number, and SLM, while Fig. 13(d)-(f) showed the visualization results of CRCM and the counting results of ID number, SLM, and CRCM. The fruit moved from right to left in the test video. In Fig. 13(a), it was clearly observed that the ID number method had a serious double counting problem. At frame 1774, the ID number count had reached 289, while only 157 fruits had been labeled according to the manually labeled ground truth file. In Fig. 13(b)- Fig. 13(c), the missed counting problem of the SLM method could be found. The SLM method was unable to count the newly detected fruits in the counted region, as shown by the yellow circles (fruits with ID 315 and 316), which led to missed counting. While the CRCM method effectively solved the problems of repeated counting and missed counting. In Fig. 13(e)- Fig. 13(f), the CRCM method accurately counted the fruits (with IDs 315 and 316) that were missed by the SLM method and was closest to the manually counted fruit count. Therefore, the CRCM method was more suitable for the passion fruit counting task compared to the ID count and the SLM method.

5. Discussion

In fruit yield prediction, target detection models play a key role in the performance of fruit yield estimation. There are significant differences in the fruit growing scenarios for various fruits, some of them simple and others are complicated. Therefore, in situations where it is not determined which target detector model is the most suitable, it is generally experimentally validated to choose the most appropriate target detector. To select the most suitable detector, this study also compared the results of the chosen YOLOv8n detector with those of the Faster R-CNN, YOLOv9s, YOLOv10n, and Deformable detection transformer (Zhu et al., 2021) detectors, and the results are shown in Table 4.

It is observed that YOLOv8n can be chosen the best detector for passion fruit detection according to accuracy and storage size. Specifically, the P, R, mAP50, mAP50-90, and size values of the YOLOv8n approach are 80.4 %, 80.4 %, 86.3 %, 49.2 %, and 6.2 MB, respectively. Compared to the YOLOv10n method, the adopted YOLOv8n can improve by 2.0 % for mAP50 and 0.8 % for mAP50-95, while only adding storage space by 0.5 MB. Compared to Faster R-CNN, the

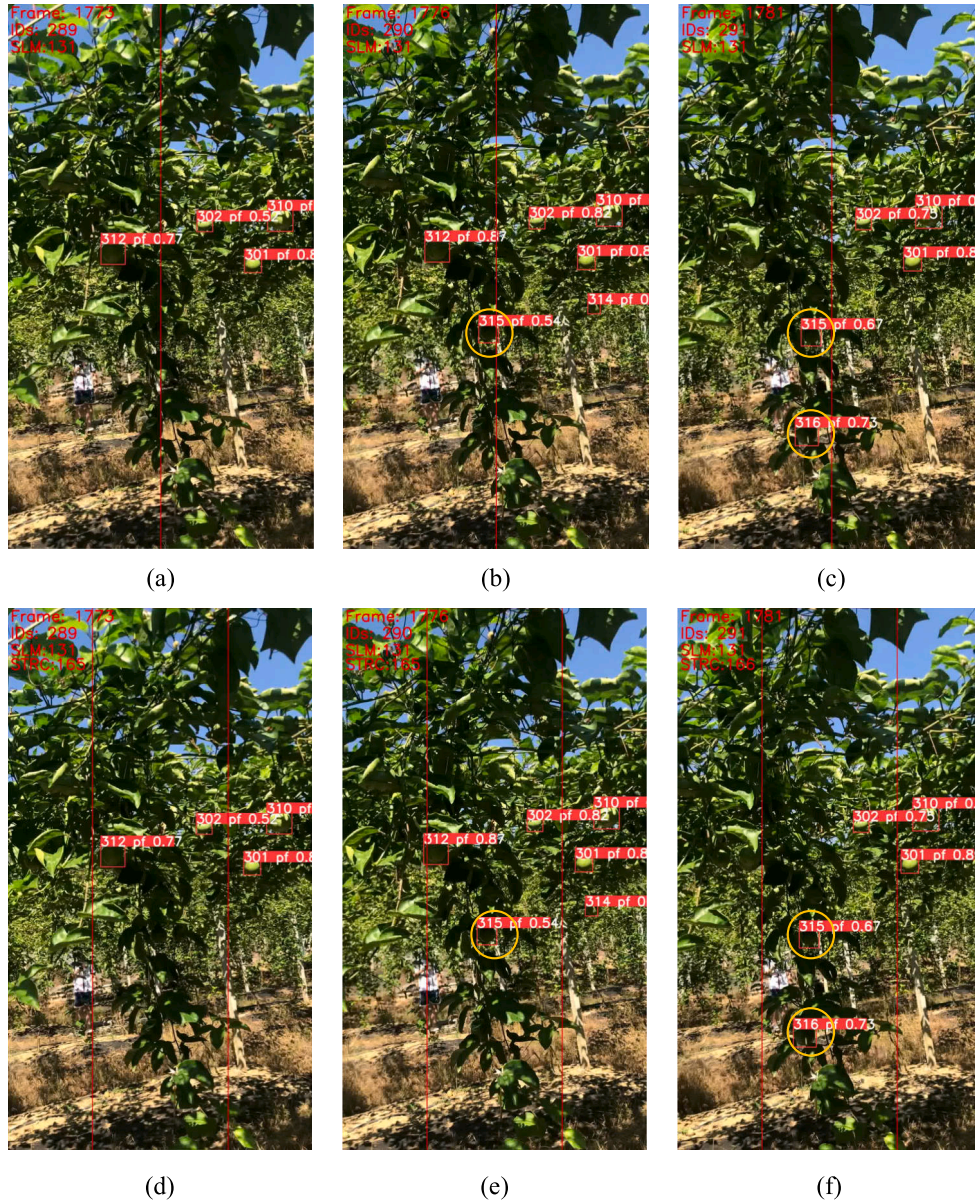


Fig. 13. The visualization results of the numbers of ID, SLM and CRCM.

Table 4

The comparison results of YOLOv8n with other detectors.

	P/%↑	R/%↑	mAP50/%↑	mAP50-95/%↑	Size/MB↓
Faster R-CNN	–	–	86.7	41.1	335.3
YOLOv10n	79.8	78.5	84.3	48.4	5.7
YOLOv9s	78.1	80.4	86.0	48.5	15.2
Deformable DETR	–	–	85.0	44.2	524.2
YOLOv8n	80.4	80.4	86.3	49.2	6.2

YOLOv8n achieves decline of 0.4 % for mAP50. The size of storage space for the Faster R-CNN is 335.3 M, which is much larger than that of the YOLOv8n. Therefore, Faster R-CNN runs slower and does not satisfy real-time requirements. Additionally, Comparing YOLOv9s and DETR methods, the YOLOv8n achieves the best performance on all metrics. These comparative results further validate the effectiveness of the YOLOv8n in fruit detection and improving the accuracy of yield estimation.

The accuracy of maximum ID number approach for fruit yield estimation is 37.2 % in our study. There may be 2 main reasons about the

inefficiency of maximum ID number method. One is that there are frequent incorrect exchanges of fruit IDs when passion fruit orchards provide a dense fruit production, resulting in low accuracy of ID counts. And our collected dataset is particularly dense and there is severe occlusion for the 1-minute video. Whereas in (Wu et al. 2023), it was shown that when calculating the number of apples in orchards, the use of removing duplicate fruit IDs and IDs of fruits lying on the ground makes it possible to achieve an average counting precision of more than 95 %. The reason may be that the passion fruit cultivation scenario differs significantly from that of the apple. Passion fruit is a vine grown graft (like the grape environment), and its fruits are particularly dense in the orchards with its very similarly colored leaves and fruits. Apples produce their fruits on a tree, and there is a significant difference in the color of the leaves and the fruit. Therefore, there is a relatively significant differences between passion fruit and apple scenarios, which can likely result in different results for the ID counting method for passion fruit versus apples.

Another reason is that the time duration of each video is varied, and there are more challenges with long video tracking. The average accuracy of ID counting method in our previous passion fruit yield estimation

(Tu S, et al.2024) is 86.19 % and it is not low. The reason may be that each video in our previous work is about 15 s in length and passion fruit orchard has sparse fruits. However, the length of each video in this study is about 60 s and dataset is particularly dense with fruits. When the illumination variations are sophisticated and mobile phone is employed for shooting, there may the camera moves quickly, resulting in unclear image condition, which caused the error ID counts.

The novelty of our work is not solely based on combining three known methods (YOLOv8n, OC-SORT, and CRCM) but on systematically evaluating and selecting the best combination of detection, tracking, and counting methods to address the specific challenges of passion fruit counting in real-world scenarios. The study identifies and implements the most effective solutions for overcoming issues like occlusion, lighting variations, and camera shaking, leading to more accurate and reliable yield estimation. This comprehensive approach, along with the creation of a real-world dataset and the performance validation against other methods, distinguishes this work from previous studies.

6. Conclusions

In this study, we actively explored the passion fruit yield estimation task in a real scenario with various challenging factors such as changes in lighting, background interference, camera shooting angles, as well as camera shake. To achieve optimal object detection performance, we conducted comparative analyses of detectors like YOLOv5, YOLOv7, and YOLOv8, ultimately selecting YOLOv8n as the most suitable detector, achieving precision of 80.4 %, recall of 80.5 %, F1 score of 80.4 %, mAP@0.5 of 86.3 %, with a model size of only 6.2 MB.

In fruit tracking, we compared tracking algorithms including OC-SORT, BoT-SORT, Byte Track, and Strong SORT. The results showed that OC-SORT performed the best, achieving 67.103 % of HOTA, 75.874 % of MOTA and 75.618 % of IDF1 respectively, ensuring more reliable fruit tracking.

Regarding passion fruit counting, we designed the CRCM algorithm and comprehensively compared with other two counting methods including ID count and SLM. In terms of average counting precision, the CRCM method performed remarkably well, achieving an average counting precision of 87.0 %, which was better than ID count (37.2 %) and SLM (76.5 %).

Considering that high fruit quantity and shooting toward sunlight may pose more challenges for passion fruit detection and tracking algorithms, such as increased mutual obstruction between fruits and frame exposure. Therefore, the utilization of the appropriate detectors YOLOv8n and tracker OC-SORT, coupled with the CRCM method, effectively resolved the issue of repetitive counting in the current passion fruit yield estimation methods. This approach enabled stable and real-time estimation of passion fruit yield. This research outcome can also be applied to similar fruit yield predictions, contributing positively to future fruit cultivation planning, farm management, and market supply chain dynamics.

CRediT authorship contribution statement

Shuqin Tu: Writing – review & editing, Resources, Funding acquisition, Formal analysis, Conceptualization. **Yufei Huang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Qiong Huang:** Supervision, Project administration, Funding acquisition. **Hongxing Liu:** Visualization, Resources, Investigation. **Yifan Cai:** Visualization, Resources, Investigation. **Hua Lei:** Resources, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

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Data availability

Data will be made available on request.

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